

Code

```
public class SelectionSort {
    public static void sort(int[] arr) {
        int n = arr.length;
        for (int i = 0; i < n - 1; i++) {
            int minIdx = i;
            for (int j = i + 1; j < n; j++) {
                if (arr[j] < arr[minIdx]) {
                    minIdx = j;
                }
            }
            int temp = arr[minIdx];
            arr[minIdx] = arr[i];
            arr[i] = temp;
        }
    }

    public static void main(String[] args) {
        int[] arr = {64, 25, 12, 22, 11};
        System.out.println("Original array:");
        printArray(arr);

        sort(arr);

        System.out.println("Sorted array:");
        printArray(arr);
    }

    public static void printArray(int[] arr) {
        for (int value : arr) {
            System.out.print(value + " ");
        }
        System.out.println();
    }
}
```

Time and Space Complexities

Time Complexity: $O(n^2)$

Space Complexity: $O(n)$